

Data Wrangling, Data Quality

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Data Quality Matters

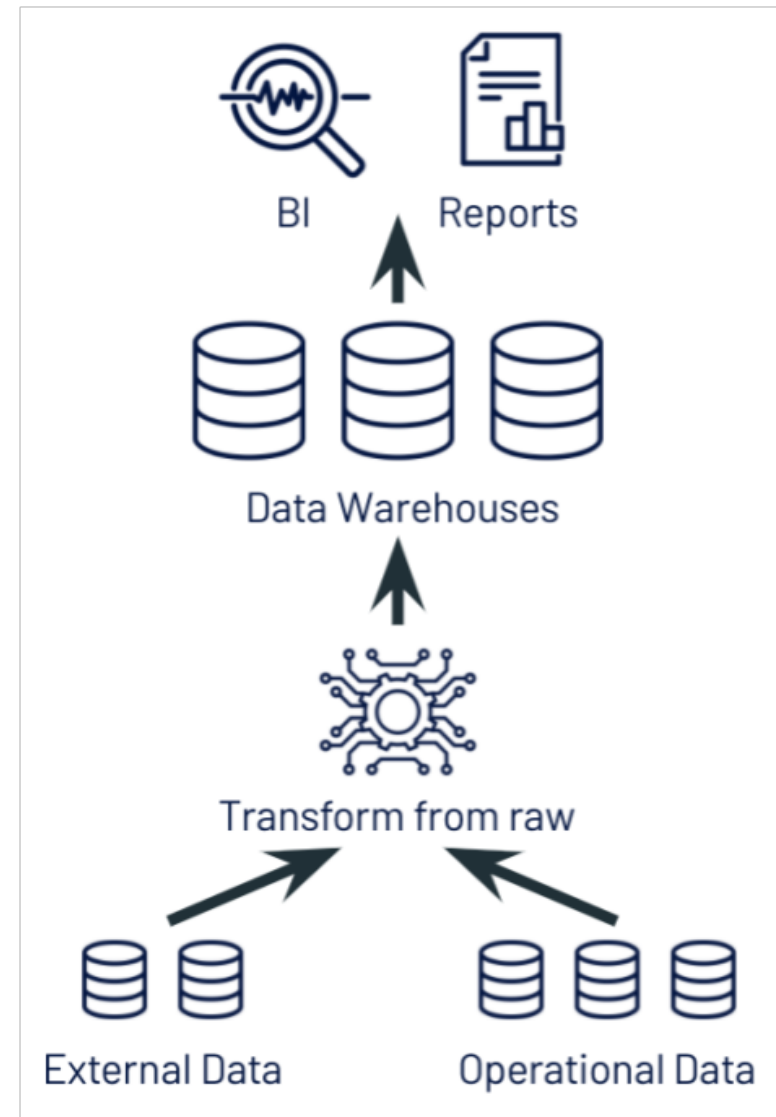
- Data quality refers to the accuracy and cleanliness of data.
 - It includes examining data consistency, completeness, and relevance.
- Without good quality data, you cannot have good analysis.
 - From entering, storing, pulling, and analyzing data, quality can become compromised at any point.
 - Data maintenance and quality assurance is an ongoing and often erroneous process in the day-to-day lives of data managers.

ETL to Deal with Data Quality Problems

- Traditionally, companies make use of ETL processes to generate and maintain good quality data warehouses that can be used for their analysis.
- EL processes are still of interest. However, they are designed to handle data that is generally well-structured, often originating from a variety of operational systems or databases the organization wants to report against.
- Large-scale data or complex raw sources that require substantial extraction and derivation to structure are not one of ETL tools' strengths.
- We therefore witnessed lately, a move away from ETL and data warehouse solution towards data lakes, where data is prepared using a variety of data wrangling solutions.

Data Warehouses

- Before the data warehouse, companies were storing data in lots of different systems with no way to unite the disparate databases.
- For analysts, a disparate system can make it impossible to synthesize a total view on which to base actionable business insights.
- Data warehouses brought the collection of databases all under a single umbrella and allowed the data to be queried and viewed as a whole.



Data Warehouses

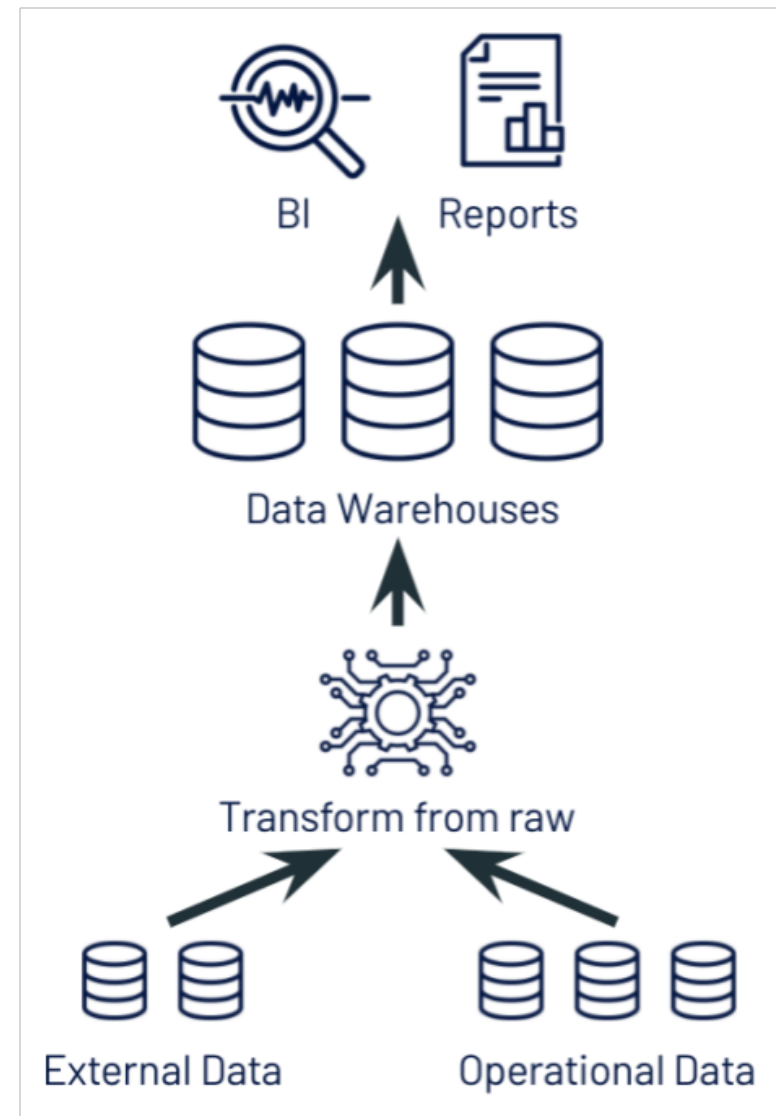
Advantages

- Standard structured query language (SQL) for access
- Integration of many data sources
- Data optimized for fast reads
- Ability to run quick ad-hoc analytical queries

Challenges

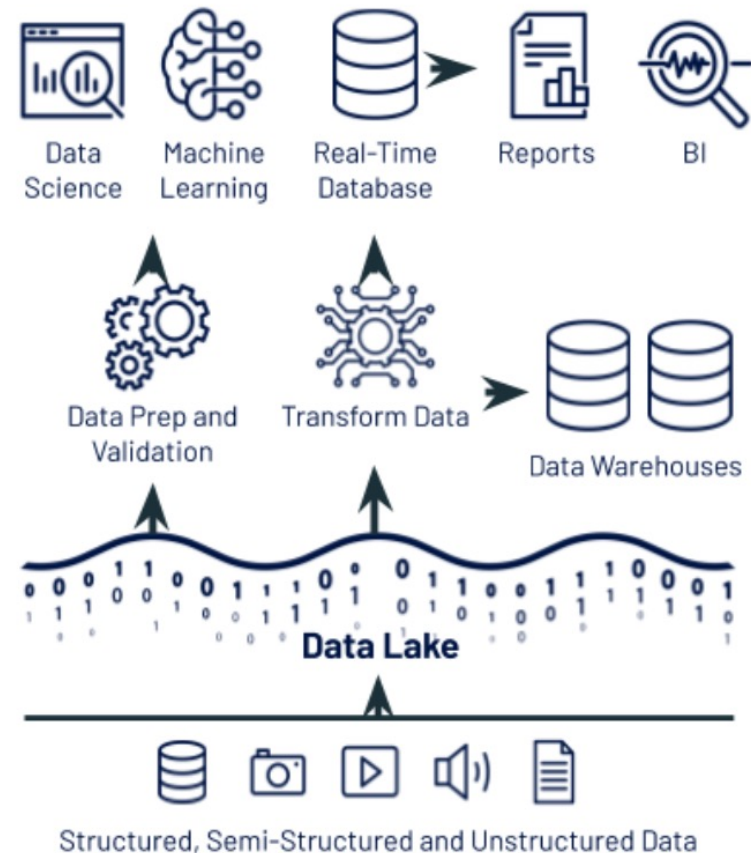
- Inability to store unstructured, raw data
- Difficult to scale
- Requires significant investment in a proprietary system

All of these add up to bottlenecks and roadblocks for data analysts.



Data Lakes

- Data Lakes emerged to answer some of the problems organizations were facing with their data warehouses.
- Data lakes are often used to consolidate all of an organization's data in a single, central location, where it can be saved "as is," without the need to impose a schema or structure on it up front.
- Data in all stages of the refinement process can be stored in a data lake: raw data can be ingested and stored right alongside an organization's structured, tabular data sources, as well as intermediate data tables generated in the process of refining raw data.



Data Lakes

Advantages

- Can hold all of an organization's data: structured, unstructured, and semi-structured
- Centralizes data for access by the whole data team
- Storage is relatively inexpensive, and the quantity of storage can be easily increased for scalability

Challenges

- Data Quality, it can be difficult to maintain data lakes which can threaten the veracity of an organizations data, in addition to managing the variety of the data
- Query performance. as data lakes grow larger, query performance suffers. This can be due to a variety of factors including slowdowns around metadata management and improper data partitioning.

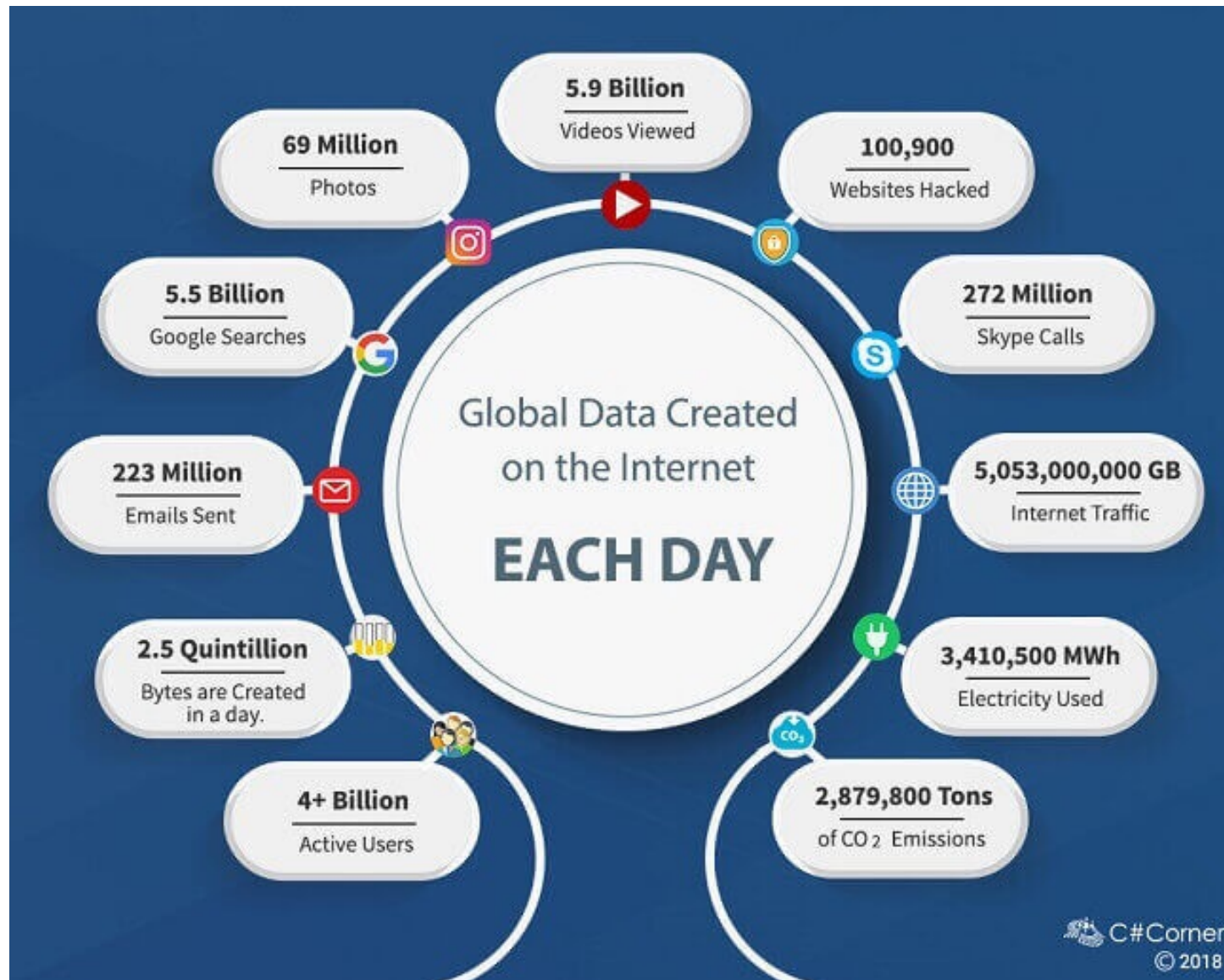
Data Warehouse vs. Data Lake (Google trend)



Data warehouse

Data lake

Volume



Source: <https://www.c-sharpcorner.com/article/how-much-data-is-on-the-internet/>

Open vs. Closed Data

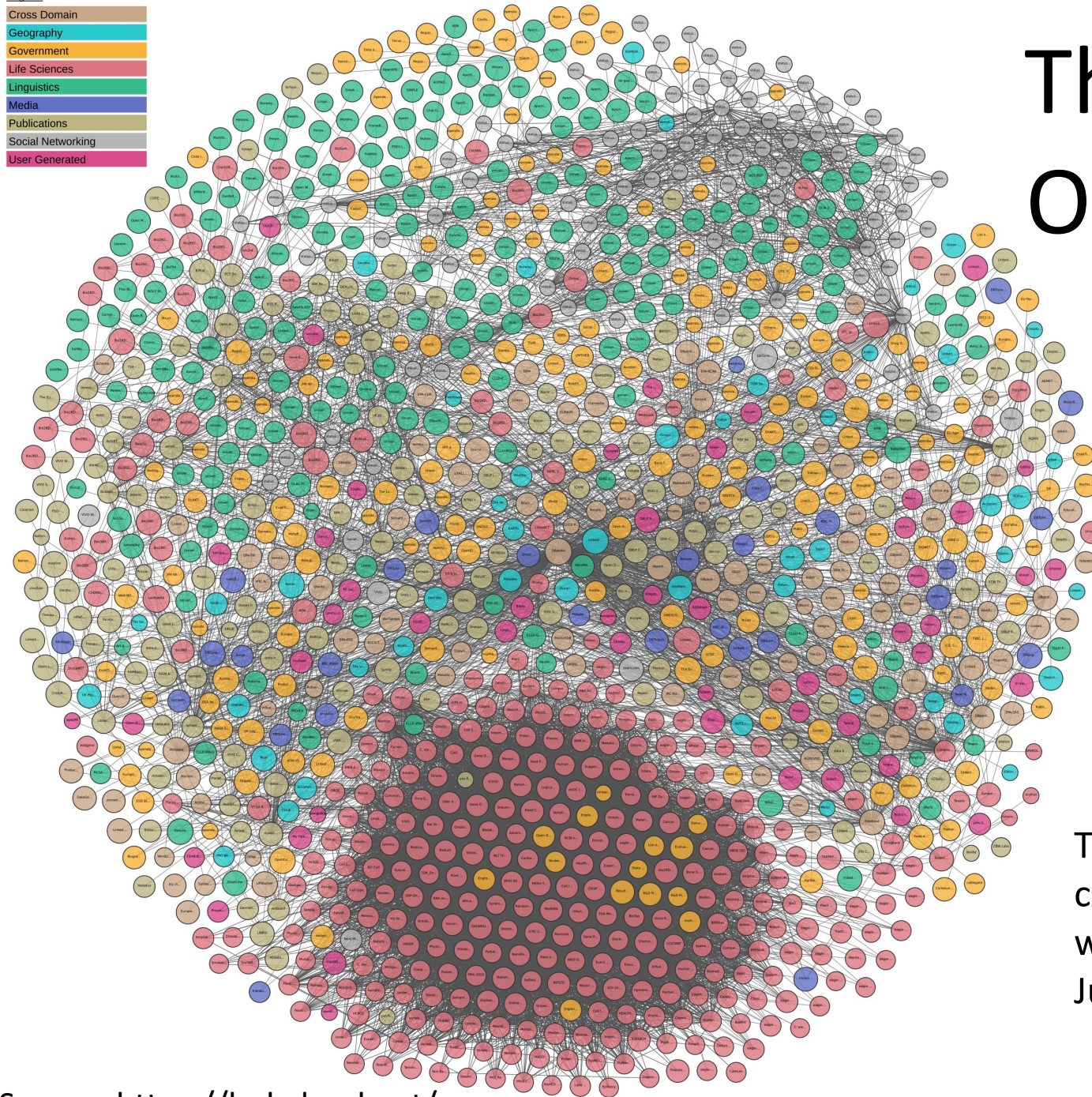
Open

- Linked data
 - <http://linkeddata.org/>
- Government data
 - data.gov, data.gov.uk
 - Eurostat
- Scientific data
 - Genes, proteins, chemicals
 - Scientific articles
 - Climate
 - Astronomy
- Published data
 - Tweet (limited)
 - Crawls
- Historical data
 - Stock prices

Closed

- Transactional data
 - Music purchases
 - Retail-data
- Social networks
 - Tweets, Facebook data
 - Likes, ratings
- E-Mails
- Web logs
 - Per person
 - Per site
- Sensor data
- Military data

Legend



The Linked Open Data Cloud

The dataset currently contains 1,231 datasets with 16,132 links (as of June 2018)

Source: <https://lod-cloud.net/>

The Linked Open Data Cloud is a project of the Open Data Commons.



Wikipedia Info Boxes

Paris Dauphine University

From Wikipedia, the free encyclopedia

Paris Dauphine University (*French: Université Paris-Dauphine*), often referred to as **Paris Dauphine** or **Dauphine**, is a selective [public research and higher education institution](#) in [Paris](#), [France](#). It is the only institution in France to be both a [grande école](#) and University^[1]. Dauphine was founded as a faculty of [economics](#) and [management](#) in 1968 in the former [NATO](#) headquarters in western Paris, in the [16th arrondissement](#).

Dauphine is renowned for its teaching in finance, economics, law, finance and mathematics and business strategy.^[2] Dauphine is a selective university with the status of "Grand Etablissement"; this unique legal status within the French higher education system allows Dauphine to make an entrance selection of its students.^[3] On average, 90 to 95% of the accepted students received either high distinctions or the highest distinctions at their French High School National Exam results (Examen National du Baccalauréat).^[4]

Dauphine is also a founding member, and now a constituent college, of [Université PSL](#). It also belongs to the [Conférence des Grandes écoles](#).

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Paris Dauphine University

Université Paris-Dauphine



Type	Grand établissement (EPSCP)
Established	1968
Affiliation	Université PSL , Conférence des grandes écoles
Budget	€115 million (2016)
President	Isabelle Huault
Administrative staff	447
Students	10,000
Location	Paris, France
Campus	Urban
Website	www.dauphine.fr 

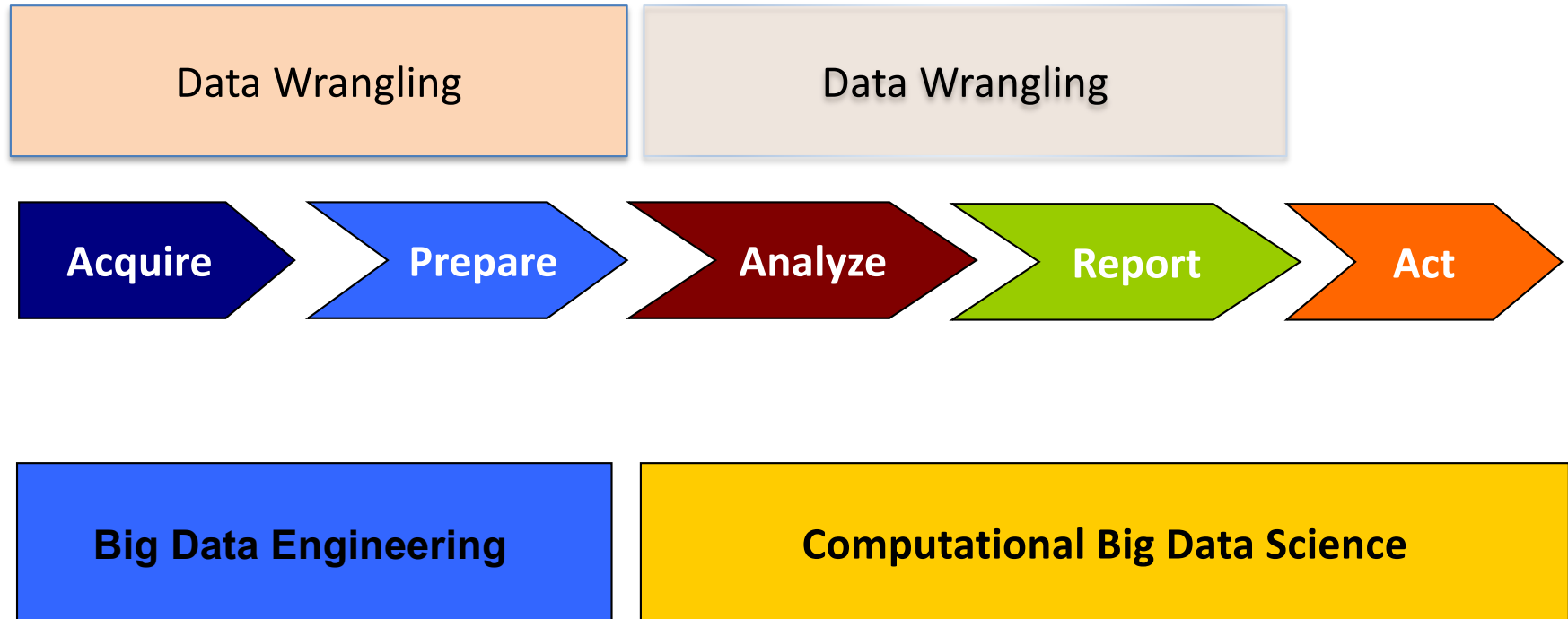
DBpedia

- From 125 languages of Wikipedia
- 3 billion triples
 - 580 million English
- English DBpedia
 - 4.6 million things
 - 1,445,000 persons
 - 735,000 places
 - 411,000 creative works
 - 241,000 organizations
 - 251,000 species
 - ...



<https://wiki.dbpedia.org/about/facts-figures>

Big Data et Process Data Science



Data Wrangling Tasks

- Data profiling
- Data cleaning: to deal with data quality problems, e.g., remove inconsistencies/errors
- Data normalisation
- Scraping: extracting information from sources, e.g., webpages, spreadsheets
- Data transformation: to get it into the right structure
- Data integration: combine information from multiple sources
- Data deduplication
- Information extraction: extracting structured information from unstructured/text sources

Take Away Message

- Data quality problem existed even before the Big Data era, but they have been accentuated due to the increase in the volume and variety (model/schema heterogeneity and data heterogeneity)
- Data Wrangling is a must data engineering skill for data scientist